



INSURANCE CHARGE ANALYSIS

**A COMPREHENSIVE ANALYSIS OF
KEY FACTORS INFLUENCING
INSURANCE CHARGES**

DATASET OVERVIEW

- **Age** → Age of the insured person.
- **Sex** → Gender of the insured person (male, female).
- **BMI** → Body Mass Index (an indicator of body fat based on height and weight).
- **Children** → Number of children/dependents covered by the insurance.
- **Smoker** → Whether the person is a smoker (yes, no).
- **Region** → Residential region of the insured (northwest, southeast, etc.).
- **Charges** → Medical insurance cost billed to the individual.



DATA PREPARATION

CLEANING STEPS

Step 1: Importing & Inspecting the data

- Loaded the dataset with Pandas.
- Explored with `.head()`, `.info()`, `.shape()`, `.describe()`.
- Listed columns and meanings.

Step 2: Data cleaning

- Checked for missing values and duplicates.
- Dropped duplicates where necessary.
- Ensured correct data types.

CLEANING STEPS

Step 3: Data Exploration (EDA)

- Found average, minimum, and maximum charges.
- Counted smokers vs non-smokers.
- Compared charges by gender, region, children.

Step 4: Data Transformation

- Created bmi_category (Underweight, Normal, Overweight, Obese).
- Created age_group (Young, Middle-aged, Senior).

CLEANING STEPS

Step 5: Visualisation with Matplotlib

- Created bar chart (average charges by region),
- Created line chart (average charges by age),
- Created pie chart (smoker vs non-smoker).

Step 6: Visualisation with Seaborn

- Created histogram (charges),
- Created boxplot (smoker vs non-smoker),
- Created scatter plot (BMI vs charges),
- Created heatmap (correlations).

VIEW THE FIRST 5 ROWS

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1338 entries, 0 to 1337
```

```
Data columns (total 7 columns):
```

CHECK DATA TYPES AND NON-NULL COUNTS

#	Column	Non-Null	Count	Dtype
0	age	1338	non-null	int64
1	sex	1338	non-null	object
2	bmi	1338	non-null	float64
3	children	1338	non-null	int64
4	smoker	1338	non-null	object
5	religion	1338	non-null	object
6	charges	1338	non-null	float64

dtypes: float64(2), int64(2),
object(3)
memory usage: 73.3+ KB
None
(1338, 7)

VIEW SUMMARY STATISTICS

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

EXPLORATORY DATA ANALYSIS

Average charge: \$13279.12

Minimum charge: \$1121.87

Maximum charge: \$63770.43

Smoker vs Non-smoker counts:

smoker

no 1063

yes 274

Name: count, dtype: int64

EXPLORATORY DATA ANALYSIS CONTD

Charges by gender:

sex

female 12569.578844

male 13974.998864

Name: charges, dtype: float64

Charges by region:

region

northeast 13406.384516

northwest 12450.840844

southeast 14735.411438

southwest 12346.937377

Name: charges, dtype: float64

EXPLORATORY DATA ANALYSIS CONTD

Charges by number of children:
children

0 12384.695344

1 12731.171832

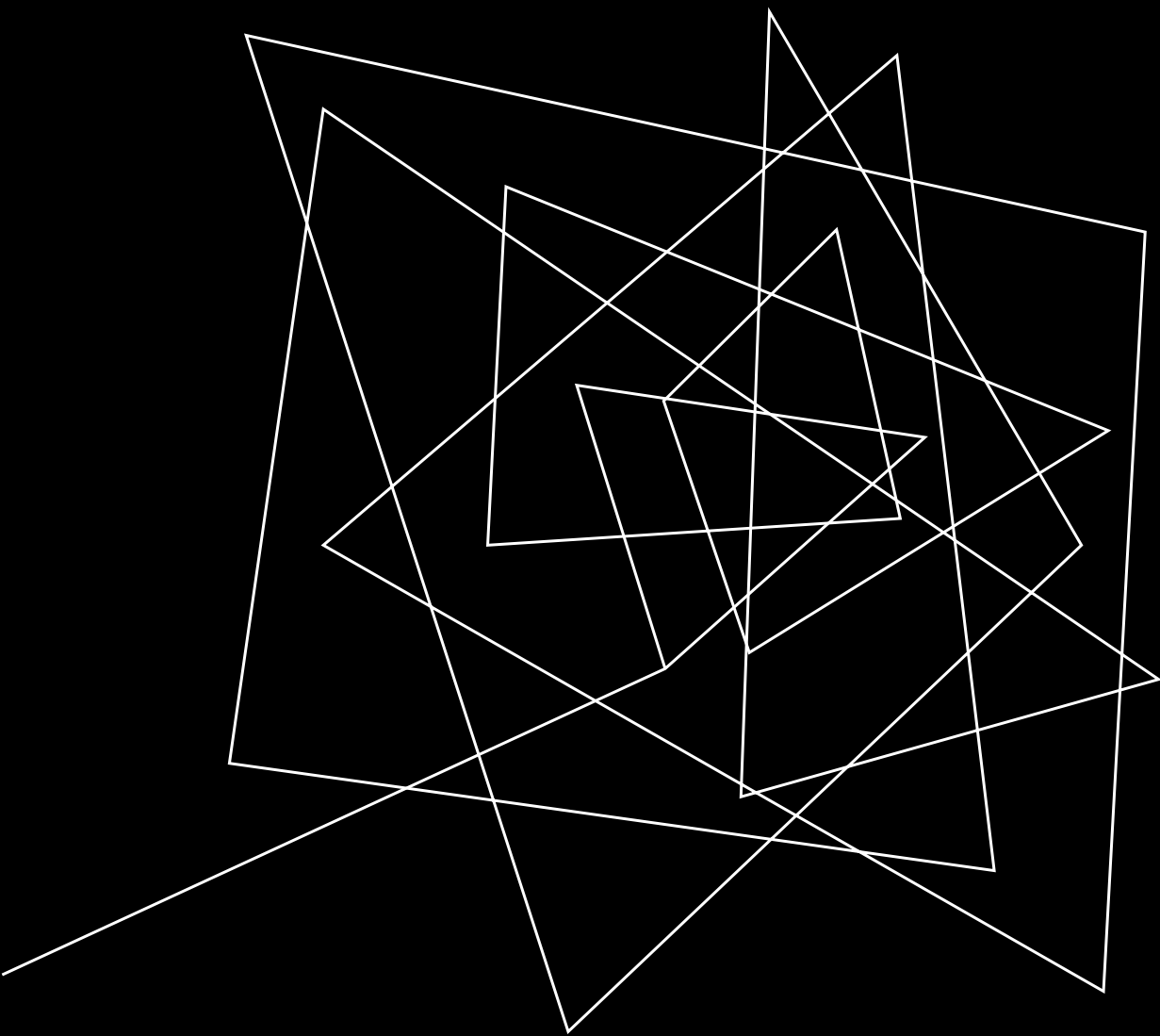
2 15073.563734

3 15355.318367

4 13850.656311

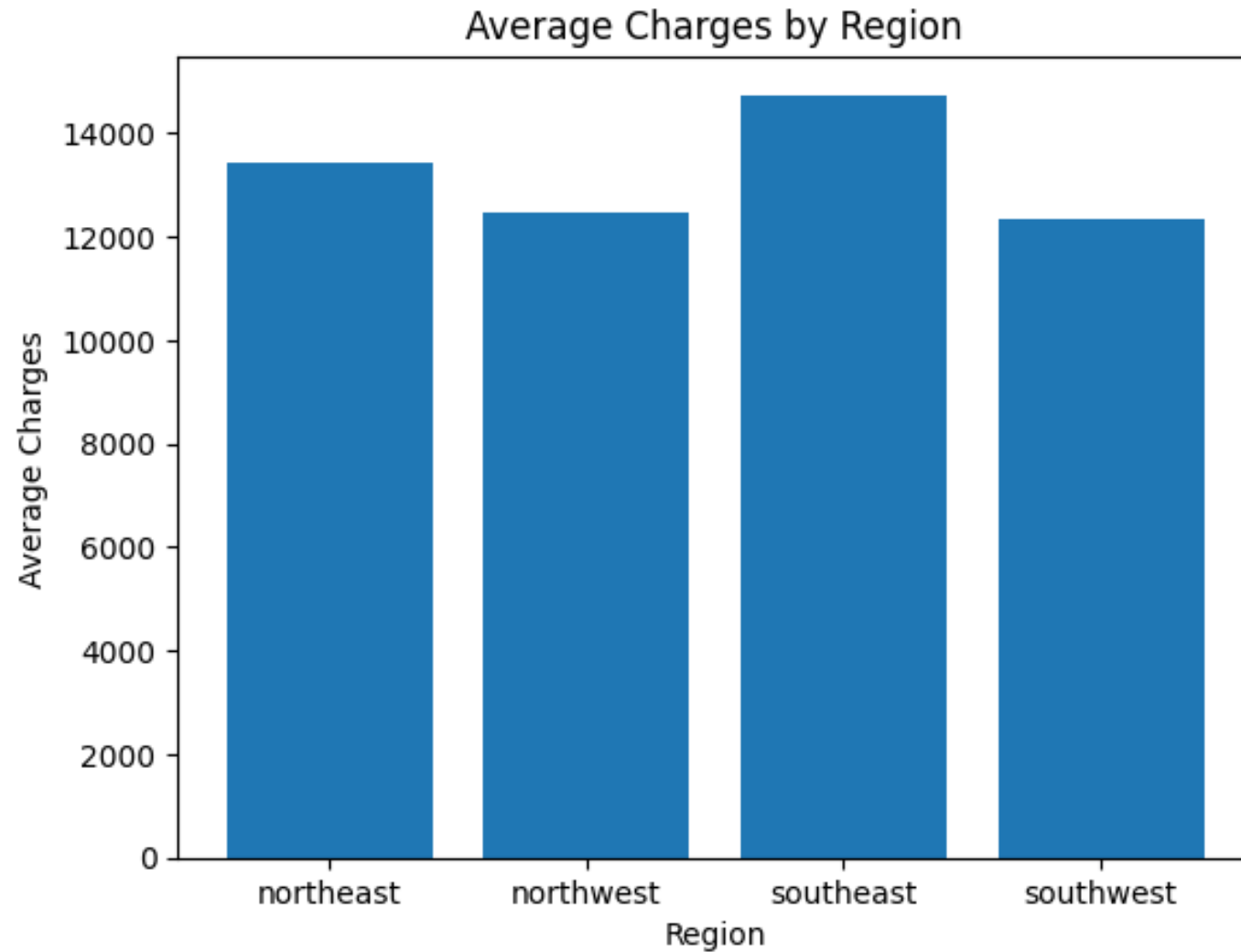
5 8786.035247

Name: charges, dtype: float64

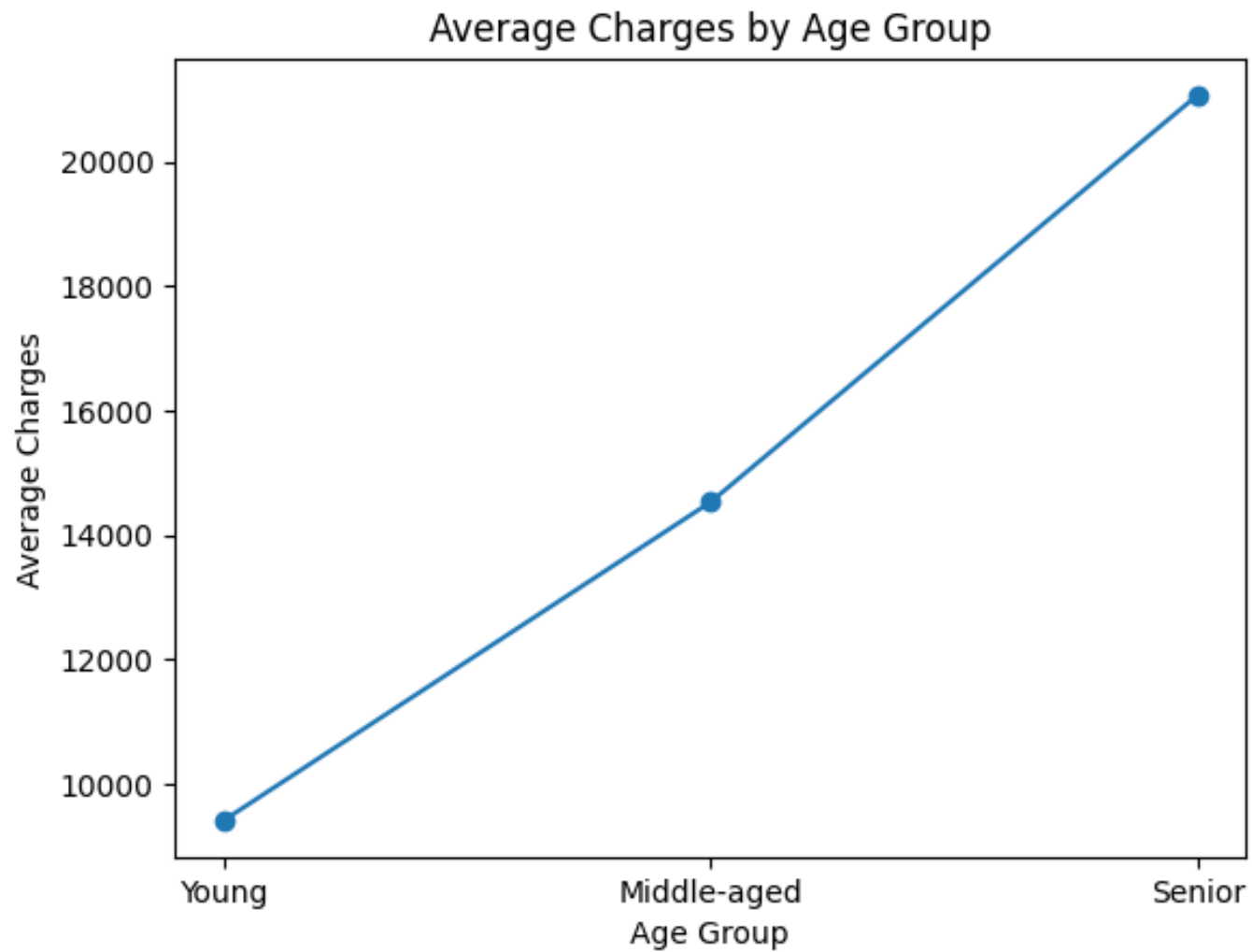


VISUALIZATIONS

AVERAGE CHARGES BY REGION

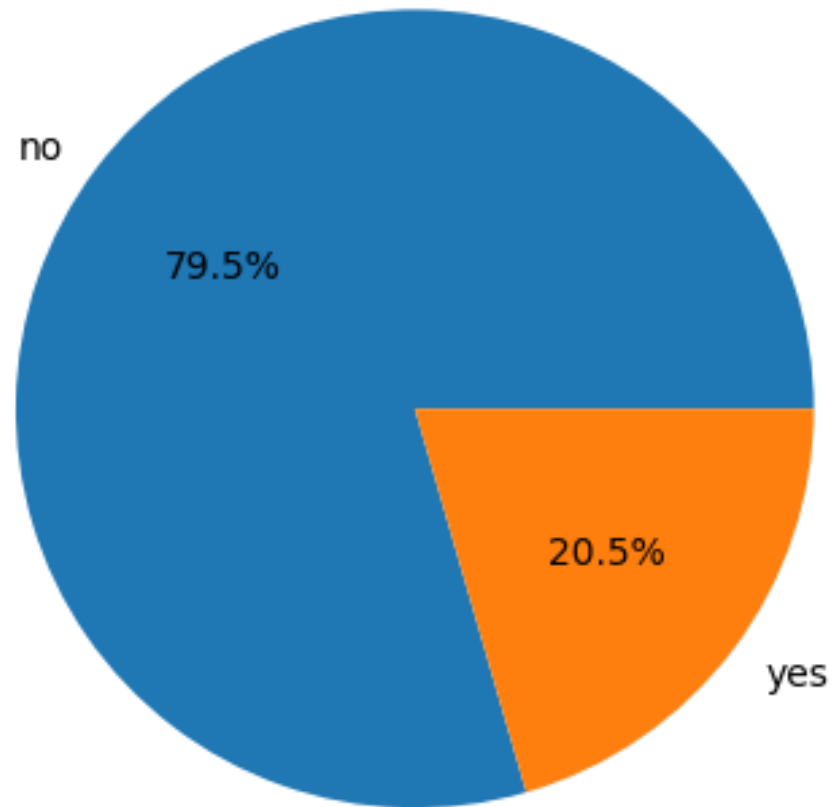


AVERAGE CHARGE BY AGE

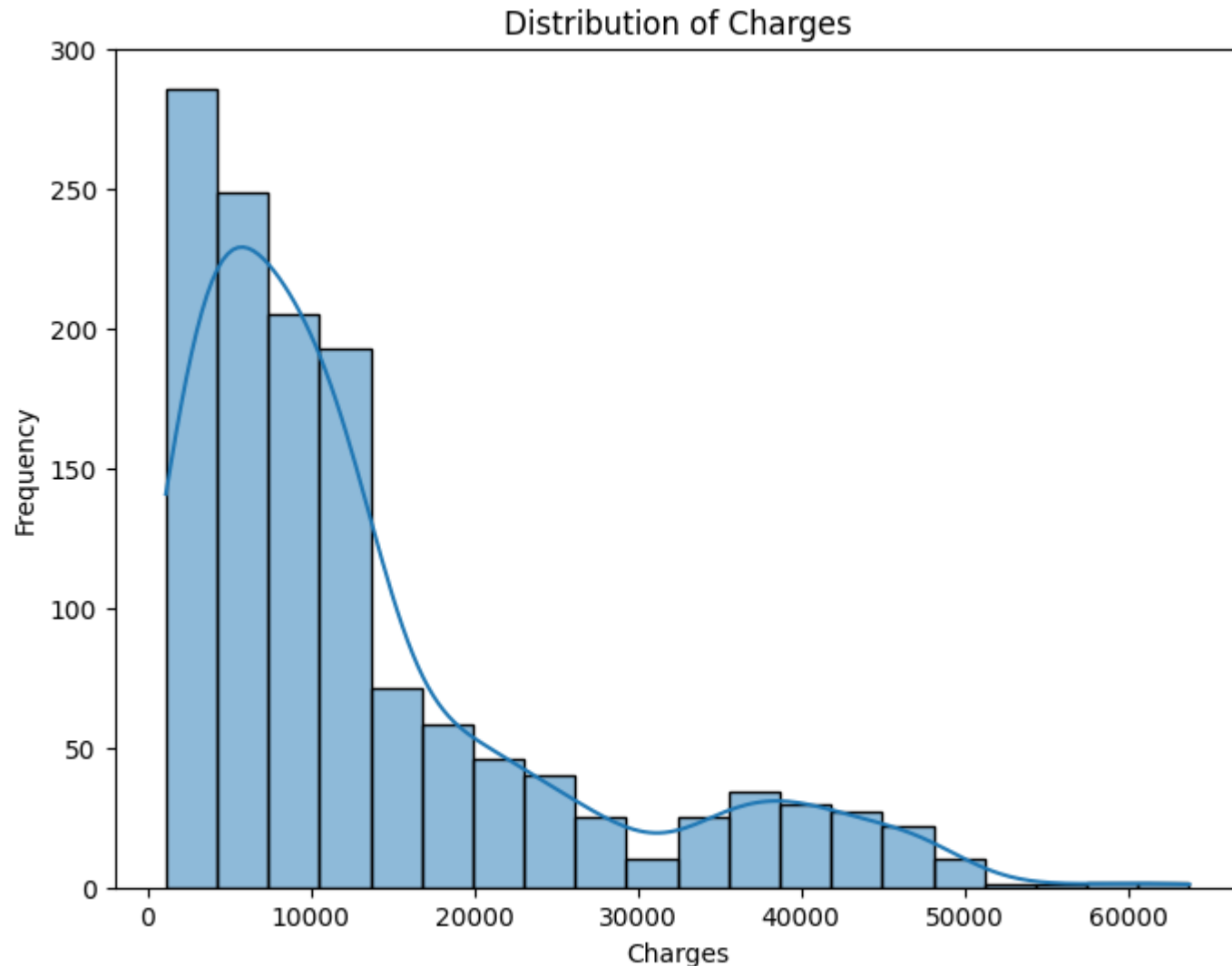


SMOKER VS NON-SMOKER

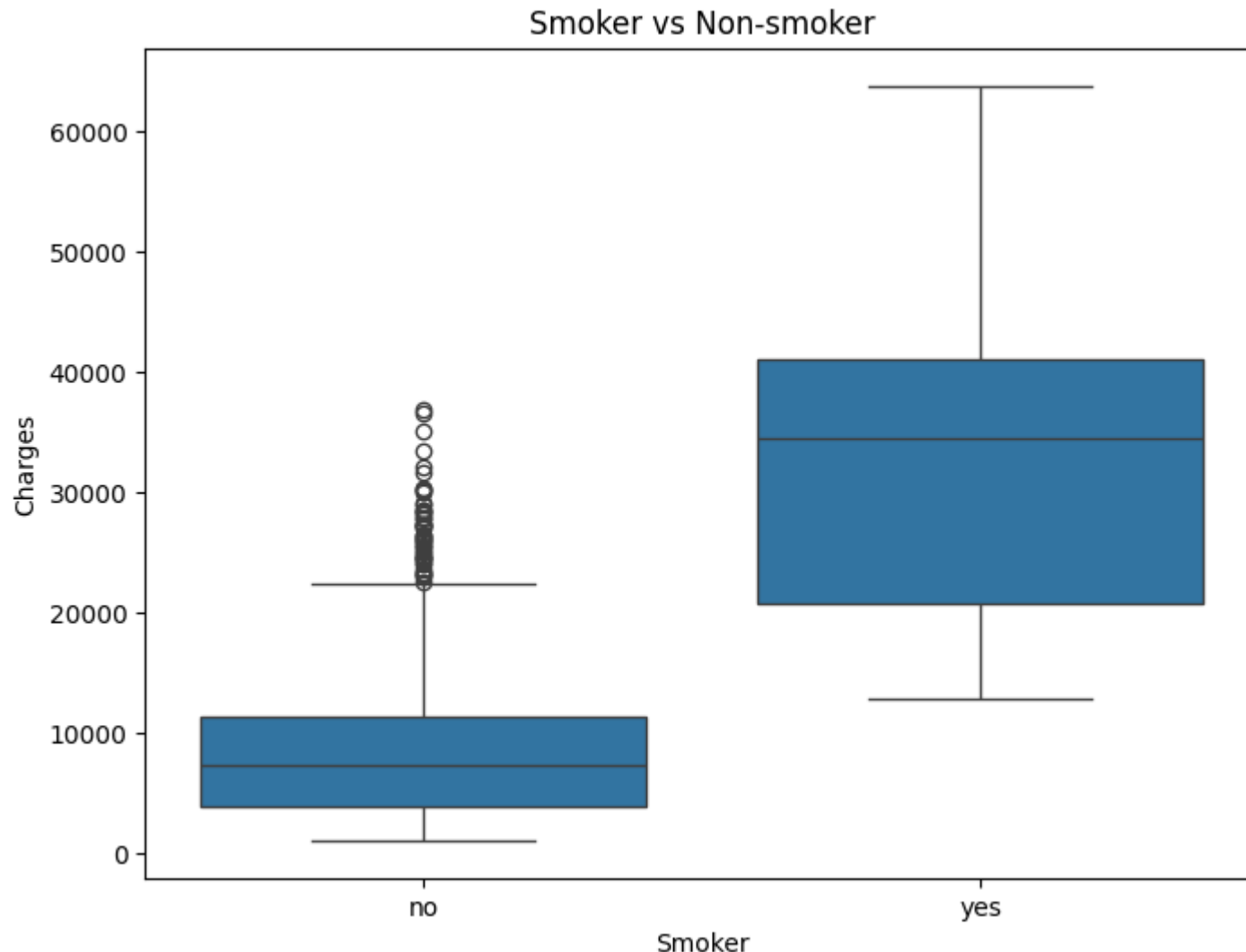
Smoker vs Non-smoker



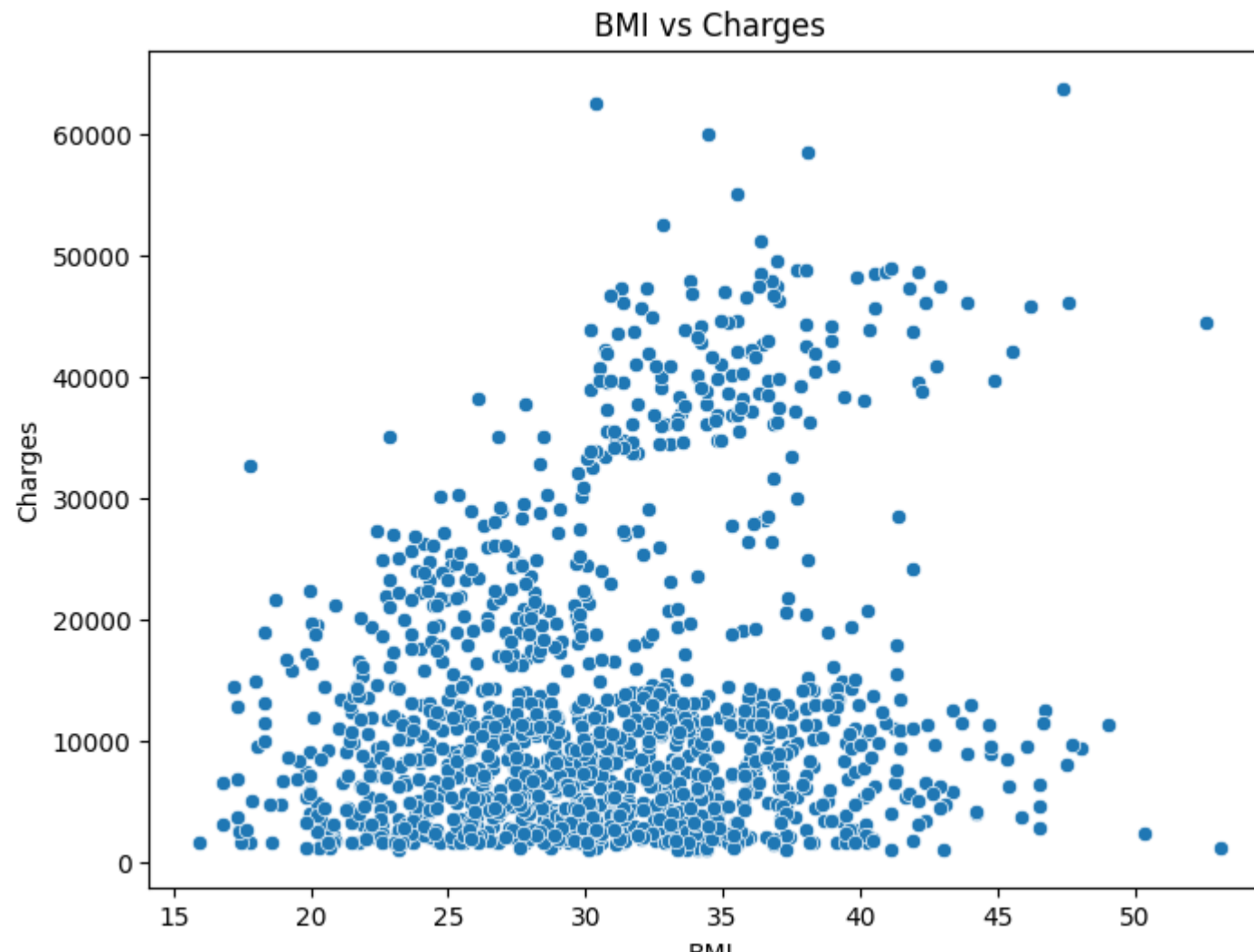
CHARGES



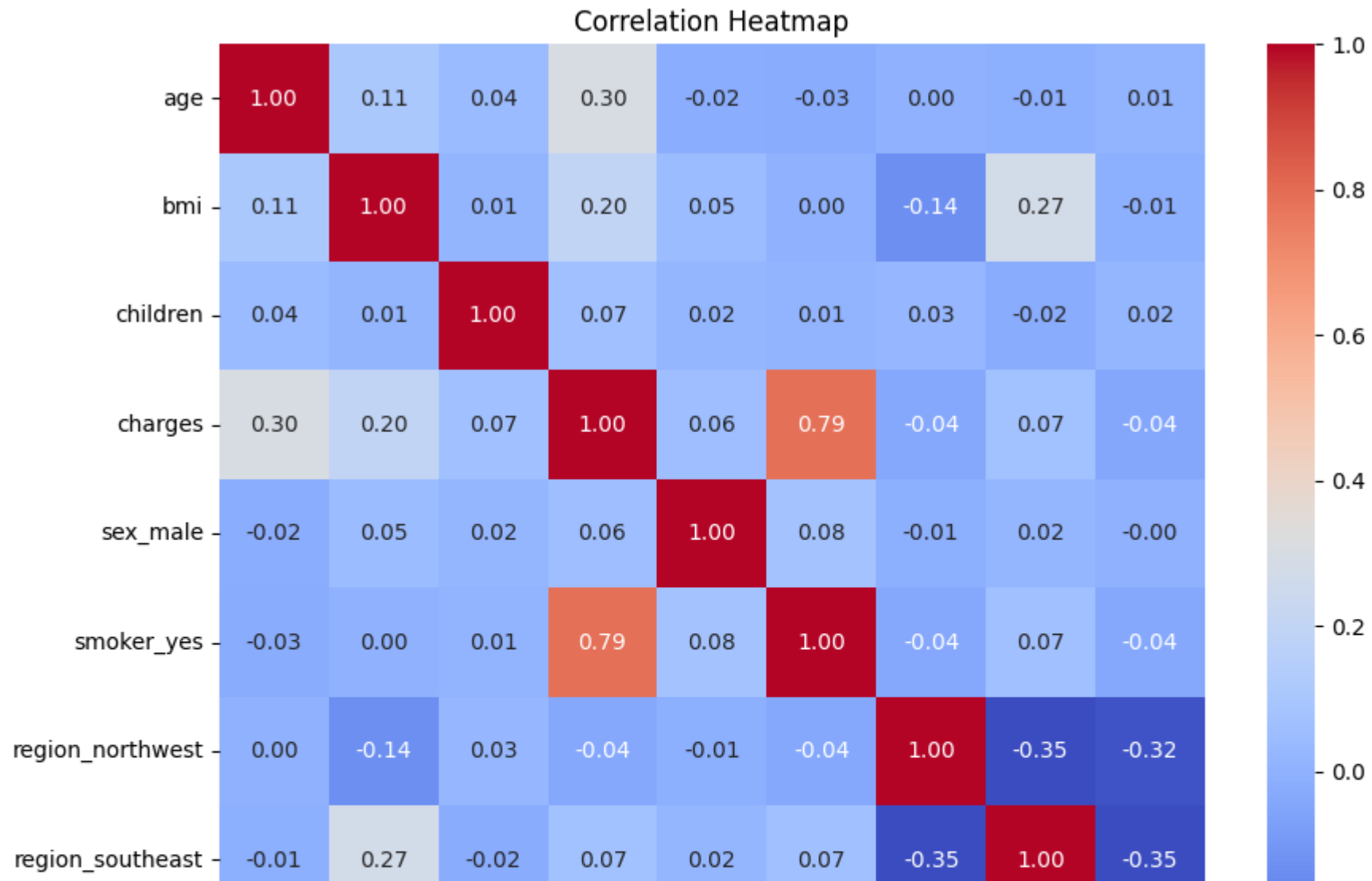
SMOKER VS NON-SMOKER



BMI VS CHARGES



CORRELATION HEATMAP



VISUALISATION EXPLAINED

Based on my analysis,

Smoking Status: Smoking Status the strongest driver of charges. The correlation heatmap shows a very high positive correlation between being a smoker and higher charges. The boxplot also clearly demonstrates a significant difference in charges between smokers and non-smokers.

Smoking is a major health risk factor, leading to various health issues that result in higher medical costs.

Age: Age has a significant positive correlation with charges. The line chart of average charges by age group shows a clear increase in charges with age, and the correlation heatmap also confirms this relationship. **As people age, they are more likely to develop health conditions that require medical attention, leading to higher insurance costs.**

BMI (Body Mass Index): BMI shows a positive correlation with charges. The scatter plot and the correlation heatmap indicate that higher BMI is associated with higher charges. **Higher BMI can be linked to various health problems like heart disease, diabetes, and other conditions that increase healthcare costs.**

KEY INSIGHTS

Smokers' vs Non-smokers: The analysis shows that smokers have significantly higher insurance charges compared to non-smokers. The boxplot clearly illustrates this difference.

BMI effect: The scatter plot of BMI vs. charges shows a general trend where higher BMI is associated with higher charges, although there's a widespread, particularly among smokers. The correlation heatmap also shows a positive correlation between BMI and charges.

Age impact: The line chart of average charges by age group shows that average charges increase with age, with the Senior group having the highest average charges. The correlation heatmap confirms a positive correlation between age and charges.

Regional differences: The bar chart of average charges by region shows some variation, with the Southeast region having slightly higher average charges compared to other regions, although the differences are not as pronounced as the impact of smoking or age.

These findings align with the correlation heatmap, which indicates that 'smoker yes' has the strongest positive correlation with 'charges', followed by 'age' and then 'BMI'.

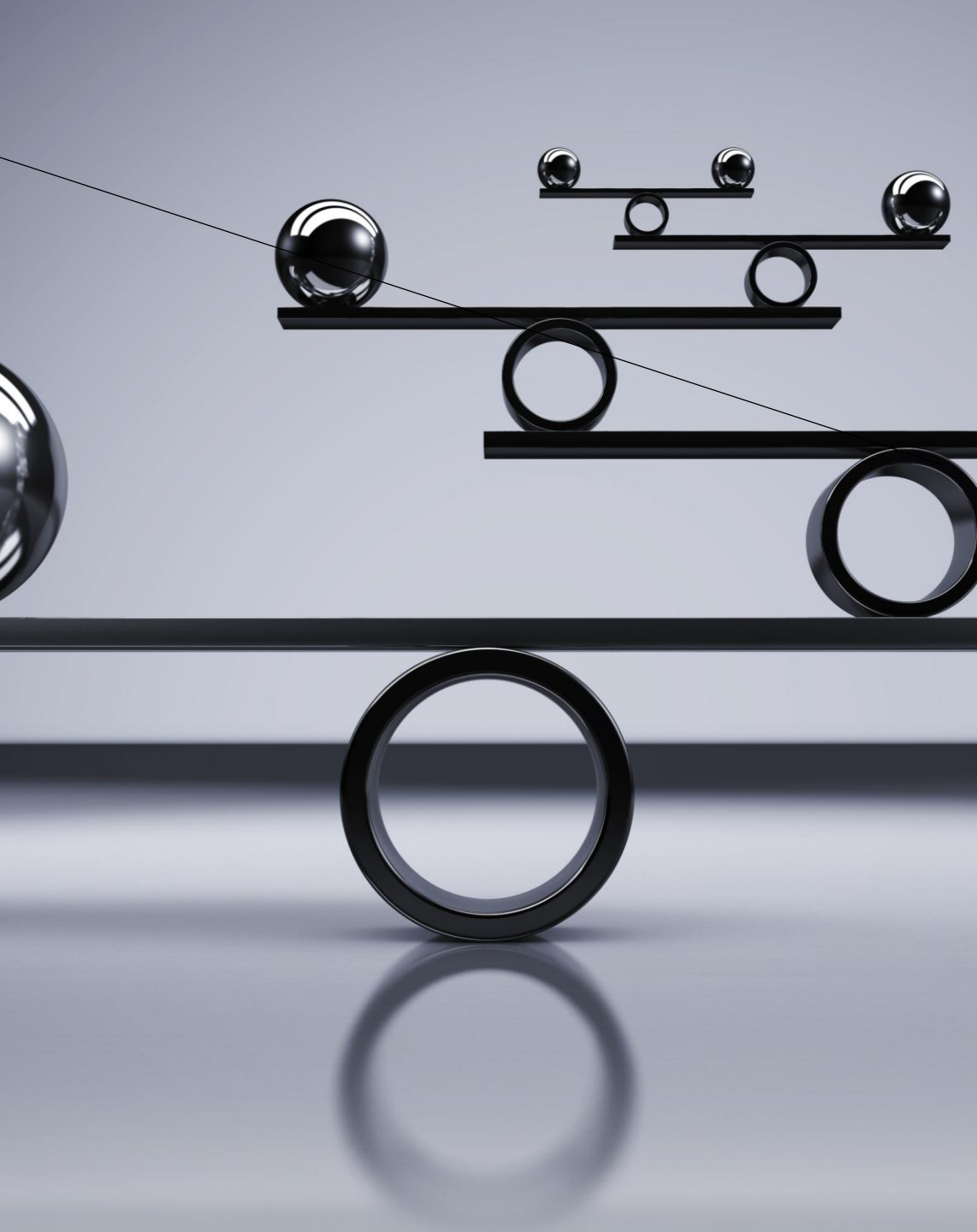
SUMMARY OF DRIVERS

Smoking Status: Smokers have significantly higher charges due to increased health risks and associated medical costs.

Age: Older individuals tend to have higher charges as the likelihood of developing health conditions increases with age.

BMI: Higher BMI is associated with increased charges, likely due to related health issues such as heart disease and diabetes.

These factors show the strongest positive correlations with insurance charges in this dataset.



CLOSING & REFLECTION

While other factors like region and number of children have some influence, **smoking status**, **age**, and **BMI** appear to be the most impactful drivers of insurance charges in this dataset.

A series of white, thin, overlapping geometric lines on a black background, forming a complex, abstract shape on the left side of the slide.

THANK YOU